SECTION

STATISTICS 20

CHAPTER

114 STATISTICS, PART 1: FIRST FOUNDATION☆

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INTRODUCTION

A good number of years ago we published a series of 37 columns in Spectroscopy magazine called "Statistics in Spectroscopy" (SIS); the series ran from the February 1987 to January 1993 issues. That series was the distant ancestor of the book "Chemometrics in Spectroscopy" that you are now reading. There is also an eponymous book "SiS" originally published by Academic Press [1] (and still available at http://amazon.com and http://barnesandnoble.com) based on that series of columns. The rationale for that series of spectroscopy columns was this: When I was first learning statistics, I was fortunate to have the consultation services of a first-class statistician available. I soon learned what could be done with data when the statistical methodology is handled properly and what statistics can accomplish when applied by someone truly knowledgeable. At that point my reaction was something like "Every experimental scientist needs to learn this stuff!" At the time it seemed that unfortunately, most chemists and spectroscopists did not have the time or inclination to learn this field of study, despite being exposed to some of the peripheral concepts and applications. This led to some incorrect and strange pronouncements appearing in the literature (sometimes, as the saying, generally attributed to the physicist Wolfgang Pauli, goes, "not even wrong!"). We published those statistics columns, and ultimately the book, as an attempt to disseminate correct statistical concepts and methodology and hopefully to motivate scientists to learn the proper use of the tools of that field. We received some positive feedback and seemed to persuade at least some scientists to learn to use statistics properly. Over the intervening years, however, the status of the field is that it has crept back to a state of authors espousing uninformed pronouncements, again providing incorrect results based on incorrect explanations. We hope to

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While a newcomer to the study of statistics can become confused and disoriented by the plethora of equations encountered, and the differences in those equations that arise when dealing with situations that are only slightly different, modern statistics rests, ultimately, on three foundational principles. So for the most part we will not use a mathematical presentation and will minimize the use of equations. Another way to express this is to say that rather than looking at the subject from the bottom-up by getting involved in the nitty-gritty of the equations, we will inspect it from the top-down, concentrating on underlying principles that can be seen to apply to all major aspects of the use and application of statistical thought.

THE FOUNDATIONS

FIRST FOUNDATION: PROBABILITY THEORY

The beginnings of the science of statistics can be traced to a famous event (famous, at least, among statisticians), wherein a nobleman was addicted to playing dice, but found that he was regularly losing money at that activity. He sought the assistance of Gerolamo Cardano, a mathematician already prominent in his day (some of the scene is dim here: How did he know a mathematician could help him? How did he know which one and how was he able to contact him? etc.) In any case, after some consideration of the nature of the problem, Cardano wrote down the 36 ways two dice can fall and counted the number of times each value (from 2 to 12) occurred. After making some assumptions about the dice (e.g., each throw of a die was independent of the other die, that the value thrown on each die occurred randomly and was independent of previous throws of that die, and that "sufficiently many" throws were made so that his conclusions were valid for the "long run"), he was able to calculate the probabilities corresponding for each value to occur. He then recommended betting strategies to the nobleman based on his findings, whereupon the nobleman started winning (over the long run, at least). We can imagine that Cardano was well compensated for his advice! Thereupon, many other mathematicians of the day, whose names are now almost household words (depending on the household, of course: Galileo, Pascal, Fermat, Leibnitz, Bernoulli, Gauss, Newton, and other scientific luminaries throughout the years) got on the bandwagon of generating and disseminating the knowledge of this new branch of mathematics. In the process of doing that, being mathematicians, they inevitably generalized and codified their results and created the theories and theorems for describing and calculating the behavior of random data, that we now recognize as the foundations of modern statistical analysis.

The next step in this saga came the realization among scientists in all disciplines, but especially astronomers at that time, that the uncontrollable fluctuations that inevitably contaminated their measurements were also caused by random factors, and could be treated similarly to the fluctuations seen in gambling, by applying similar methodology to the scientific data as was applied to gambling results.

In modern times, of course, we recognize that virtually any and all measurements are subject to fluctuations from uncontrollable sources. We generally call these fluctuations "noise" or "error." Metrologists, such as those at NIST, go further and recognize that errors can be of two types, systematic and random, and designate the total departure of a given measurement from "truth" as the "uncertainty" of the measurement. Systematic errors are presumed to arise from mechanical, electrical, optical, or other physical causes. Then by application of knowledge from the appropriate field, the source of the discrepancy can be tracked down and eliminated. For this reason, the science of statistics, and

perhaps, better understand the operation of the algorithms by relating them to the basic principles presented here.

Both of these principles make use of properties that all data have, even data that include a random contribution. Indeed, in some situations the random component of the data plays a key role in the formulation and application of the basic principle. Both principles are characterized by an (sometimes unexpected) underlying mathematical equality that exists between apparently different quantities. These two foundations will be discussed in forthcoming chapters.

REFERENCE

[1] H. Mark, J. Workman, Statistics in Spectroscopy, second ed., Academic Press, New York, 2003.

CHAPTER

STATISTICS, PART 2: SECOND FOUNDATION☆

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CHAPTER OUTLINE

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SECOND FOUNDATION: THE PRINCIPLE OF ANALYSIS OF VARIANCE

A reader of chemometric literature intending to comprehend the operation of the algorithms used to analyze spectral data will encounter an almost mind-numbing plethora of terms, abbreviations, and acronyms: R, R^2, RMS, RMSD, SEC, SEE, SEP, etc. Upon reflection, the reader must soon wonder, beyond "what do these all mean?" (which is usually explained) to other questions "why do we use THESE terms?", "where do they come from?", "what do they tell us?", and deeper questions of that nature (which often aren't explained). The answer, which one might guess from the title of the chapter this issue is brought up in, is analysis of variance (often abbreviated ANOVA).

So what is ANOVA and what does it have to do with the use of terms like the above?

Let's start here: "Everyone knows" that the spread of data is usually described by the Standard Deviation (SD) of that data. There are several good reasons for this.

- The computation of SD used to be somewhat onerous when done by hand, but with computers, and even with some hand calculators available nowadays, the computation can be done with the press of a button; some hand calculators even include that computation among their basic functions. (One might wonder why that is so.)
- The SD gives the spread of the data in the same units that the data is represented in.
- The SD automatically adjusts the measure of the spread to the actual spread that the data represents.

Statisticians know all that and also know some deeper reasons for using SD as the measure of data spread. They also like the method well enough and find it a useful tool just as everyone else does.

Statisticians, however, LOVE variance (which is the square of the Standard Deviation).

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The general approach that is used for all hypothesis tests is to say that if the two means are "very close" to each other, then we conclude that the observed difference can be attributed to the random fluctuations inherent in the data. If the difference between the means is much larger than can be attributed to the inherent random fluctuations, then we much conclude that some other effect is operating, in addition to the inherent random fluctuations, to change the value of one or both means; then we can say that the difference is statistically significant.

This *t*-test is very straightforward and is taught in many nonstatistics courses, in advanced courses in other sciences, for example. But it also has a limitation: What if there are more than two samples' worth of data? Then the simple *t*-test does not suffice. Here's where the more powerful technique of ANOVA can be applied.

ANALYSIS OF VARIANCE OF SYNTHETIC DATA

If we look at Table 115-1A we see four columns of synthetic data, each column containing 20 readings. To make the example concrete, let's imagine that these data represent measured values of responsivity of four IR detectors; with each detector having had its responsivity measured 20 times. At the bottom of Table 115-1 we also see the computed values for some statistics of those data: their mean, SD, and variances of each detector's data. We note that the means are not exactly the same for all four detectors and neither are the SDs. We can attribute this to the effect of the inherent variability of the data on the computed values, as we described earlier. But then the question arises of whether the means are different enough from each other to indicate a possibility that the differences between these four means are larger than the random contributions to the data can account for? Here is where ANOVA can help us, and here's where things get a bit tricky.

We can write the total variance of the data, for the entire data set in Table 115-1A as the sum of two contributions:

$$\text{Total variance (TVar)} = \text{Variance due to differences between the means of the four data subsets (between detectors : BVar)} + \text{Variance of the data of each detector around its own mean (within each detector : WVar)} \tag{115-1}$$

The corresponding SD we'll designate SDB and SDW.

The actual total variance is not important to us here, or to our calculations, and so will not appear in any calculations. What IS important, however, is that the two variances have another relationship between them, which will become clearer when we know a little more about what the two terms signify.

SDB is simply the SD of the four means. The "B" tells us that it's the SD "between" the samples.

SDW is a bit more complicated to explain (and to compute) but is a measure of the SD of the data. It's what we get when we combine the four Standard Deviations from the four detectors into a single SD that best represents the variability of all the detectors. Statisticians refer to this action as "pooling" the data values. The calculation is the RMS of the SDs from the four detectors. The "W" tells us that it's the variance "within" each detector's data, that are combined to give a single overall value for all the detectors.

To keep terminology simple, we convert long names such as those that are listed earlier in Eq. 115-1 to abbreviations; this is common statistical practice. We therefore express Eq. 115-1 as follows:

$$\text{TVar} = \text{B(etween)Var} + \text{W(ithin)Var} \tag{115-2}$$

Dividing one of these values by the other is called an F test, because ratios of variances follow what is known to statisticians as an F distribution:

$$F = 0.1219/0.0671 = 1.81$$

Well, that seems like an appreciable difference. But is that difference large enough for us to say that there is statistical significance to that difference? The way to answer that sort of question is to compare the F-value you calculated from your data to the known threshold F-values that statisticians have calculated for all sorts of experimental and statistical conditions and compiled in tables; alternatively, some computer programs have the capability of computing these threshold values "on the fly" as the data is analyzed. These values define the threshold (what statisticians (and we ourselves have defined earlier) call the "critical" value) of the test statistic, beyond which the results are considered statistically significant. For our example, the critical value is 3.13. Since the value we obtained from our data (2.10) is less than the critical value, we have no evidence for thinking that any effects other than the random variability of the data are operating to cause the spread seen in the means.

THE EFFECT OF ADDING A PERTURBATION

In Table 115-1B we have a similar set of data. In fact, Table 115-1B is the same as Table 115-1A except that a small constant (0.2) has been added to all the entries in one of the columns and subtracted from all the entries in one of the other columns. This is to simulate a small physical effect operating on those two detectors, to change their responsivities. We can now go through a similar analysis of Table 115-1B as we did for Table 115-1A except that we omit the extended explanatory comments.

From the data in Table 115-1B, we can calculate SDB and SDW:

SDB = 0.4804
Pooled SDW = RMS = 1.1591 (the same as for the original data, since we haven't changed the relationships between the several readings from each detector)

Converting these to variances, simply by squaring them:

Variance (Between) = BVar = 0.2308
(Pooled) Within variance (WVar, also termed MS) = 1.3435 (also the same as previously)
$1/n$ * WVar = 1.3435/20 = 0.0671 (also the same as previously)

And finally:

$F = 0.2308/0.0671 = 3.43$

Since the conditions are the same as for Table 115-1A, the (tabled) critical value of F for Table 115-1B is the same as for Table 115-1A: 3.13. We see, therefore, that the F-value for the measured data is now greater than the critical value so that we know that even though our perturbation was far smaller than the SD of the data, there is evidence of some real physical difference between the various detectors. This also illustrates how adding variance increases the total variance. ALL sources of variance add, even those due to artificial, external constants. The ANOVA by itself does not tell us which detectors are the ones causing the data to show statistical significance nor does it, by itself, tell us how much perturbation there was to the data; further measurements and experimentation would be needed for

Eq. 115-3 can be applied, of course, to each sample in Fig. 115-1. It is too complicated to do this here (the interested reader can find a derivation in [1]), but if we add up the corresponding terms for each sample, we can apply ANOVA to the differences shown in Fig. 115-1, and we find that:

$$\sum_{i=1}^{n} D^2 = \sum_{i=1}^{n} A^2 + \sum_{i=1}^{n} B^2 \tag{115-4}$$

where n is the number of samples.

These various summations and relations between them, augmented with auxiliary constants, mathematical operations, or other information in the data, become the various statistics we mentioned at the beginning of this section.

For example (note that all summations are taken over the n samples comprising the data set):

$$SEC = \sqrt{\sum B^2/(n-m)} \tag{115-5}$$

(where n is the number of samples, m is the number of PCR factors or MLR wavelengths used in the model)

$$R^2 = \frac{\sum D^2 - \sum B^2}{\sum D^2} \tag{115-6}$$

This is just one small application of the use of ANOVA, to a situation that many readers will be familiar with. Many of the statistical and chemometric operations we perform have this fundamental technique underlying them, since they are based on computations from similar pieces of the variance.

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[1] H. Mark, J. Workman, Statistics in Spectroscopy, second ed., Academic Press, New York, 2003.

CHAPTER

STATISTICS, PART 3: THIRD FOUNDATION☆

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CHAPTER OUTLINE

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THIRD FOUNDATION: THE PRINCIPLE OF LEAST SQUARES

As we described in Chapter 114, probability theory was first developed around the mid-1600s, with the original primary application being the description of games of chance (gambling), later on being applied to scientific measurements that contained a random component (noise or error). Scientists, of course, were interested then, as they are now, in determining the systematic relationships between various quantities. Mathematicians had the luxury of dealing with theoretical relationships which, whether a straight line or some curved, nonlinear relationship, were always exact, since the underlying *gedankenexperiments* describing the systems under study were always noise free. Experimental scientists, and I will pick on the astronomers here, did not have that luxury, they were stuck with whatever data they measured, which usually contained at least some noise. This led to difficulties. For example, as most if not all of us now know, an important question back then was whether the orbits of the planets were circles (which were considered to be the "perfect" shape and therefore of divine origin) or some other shape, such as ellipses. In an ideal error-free world (er, universe) it would only be necessary to ascertain the position of a planet at four different times (or five times if the measurements could not be made at equally spaced time intervals) in order to determine whether the orbit was circular or elliptical.

As all of us now know, however, measurements are not without error. All measured values have some error associated with them, whether or not we know what that error is (as was once pointed out to me, however, if you actually know the error of a particular measurement, then the quantity itself could be known error free, since you could make a mathematical correction for the error of that measurement and get an error-free final result). So when we say we know the error of a measurement, what we really mean is that we know the distribution of random values that affects the value of the underlying quantity, which is presumed to have some "true" value that we want to ascertain. So the question the mathematicians asked themselves was "How can we determine, from our measurements, what the

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very well what we stated earlier: all physical measurements inevitably are contaminated with noise and error. So how could he then make such a statement? Clearly the answer is that Gauss knew what we all now know, that zero error, being a physical impossibility, is not a hard requirement on the data. The practical meaning of that requirement on modeling is that the error must be small enough to be a good approximation to zero. Or, to put it in statistician's language, the error must be small enough to be a good estimator of zero. Then if the data meet that requirement, good modeling performance (i.e., reliable analytical accuracy) can be obtained. In modern spectrometers, the noise level can be as small as 10^{-6} compared to the signal level. 10^{-6} is a pretty good approximation to zero, for the vast majority of situations of practical interest. Detector noise is a limiting noise source. While detector noise is not the only noise source in a measurement, it is rare for any other sources of noise to be considered.

In the early days of NIR analysis, "sample noise," in the form of what has come to be loosely called "particle size effect" or "repack effect," was recognized and attempts made to measure its magnitude [3,4] (also using ANOVA). It quickly became apparent that the calibration algorithm, by virtue of their least square property, created difficulties in this endeavor. Those spectral variations due to repack that can be accommodated by the calibration and thus do not generate differences in the predicted values of the analyte obviously cannot be measured. Only those spectral variations that cannot be accommodated by the calibration algorithm and thus do generate differences in the predicted values of the analyte can be measured. Those variations, appearing only in the analyte values, are in general indistinguishable from variations due to the random detector noise and, we hope, are no larger than the variations due to the random detector noise.

And that is why modern analytical spectroscopy works.

From the above examples we can also see that the concept of least squares joins the concepts of probability and of ANOVA as the fundamental operations that underlie several of the most common and important operations that we perform.

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CHAPTER

117 HOW TO SELECT THE APPROPRIATE DEGREES OF FREEDOM FOR MULTIVARIATE CALIBRATION

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INTRODUCTION

The question of how to establish the appropriate number of degrees of freedom within the calibration and validation process when using multivariate equations varies with the person, committee, application, and interpretation where calibration and validation are used. There have been ASTM International, USP, and ISO committees attempting to standardize the mathematics and approaches; however, disagreements and discontinuities still exist. This chapter will address some of the questions and remaining issues related to degrees of freedom used for calibration and validation in spectroscopy [1–4].

Pierre Dardenne posed two questions related to chemometrics in a Spectroscopy paper [4] as follows:

> "(1) With all the stats using independent test sets (SEP, RMSEP, RMSECV, …), why do you use the parameter k or $k-1$ in the degree of freedom? Usually the sum of squared deviations is divided by N only, for sample sets not included in the modeling.
> (2) When testing a model with 1 independent test set, the approach is ok. But when comparing different models with the same sample test set, I believe the test must be a paired test. Tom Fearn presented this approach years ago in reference [5]."

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THE SUBJECT OF DEGREES OF FREEDOM

To answer these questions we pose the following discussion related to the SEE for predictions using simple linear regression or MLR. By statistical convention the degrees of freedom (df) for simple and multiple regression refer to the number of independent samples (N) minus the number of calculated values used to formulate the standard error parameter computation. For example, computing the SEE for a simple linear regression would use $N-2$ for df, designating the number of independent samples (N), minus k, represented by one df for the slope calculation (i.e., regression coefficient) and one df for the intercept calculation. Eqs. 117-1a and 117-1b illustrate this point as the SEE for simple regression. Note that Y_i is the reference value for the ith sample and $\hat{Y}_i$ is the predicted value for that same ith sample using the regression model.

$$\text{SEE} = \sqrt{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 / N - k} \tag{117-1a}$$

$$\text{SEE} = \sqrt{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 / N - 2} \tag{117-1b}$$

For multiple regression, the k technically also represents the number of regression coefficients calculated for the degrees of freedom, with an additional degree of freedom for the intercept calculation. Thus the degrees of freedom would be k, as the number of factors used in the multiple regression, and one additional df for the intercept calculation. This is illustrated by Eqs. 117-2a and 117-2b when *nine* factors (as an example) are included within the regression model.

$$\text{SEE} = \sqrt{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 / N - k - 1} \tag{117-2a}$$

$$\text{SEE} = \sqrt{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 / N - 9 - 1} \tag{117-2b}$$

For simple multiple regression, the df regression $= k$ (i.e., the number of predictors or independent variables, namely the slope and intercept) and for multiple regression, the df $= N - k - 1$, where k is the number of factors, regression coefficients, or latent variables. It is elementary to observe that a more standard nomenclature might just as easily be represented as Eq. 117-2a, where $k =$ the number of regression coefficients (slope or factors) and 1 represents the df for the intercept calculation. Note that this reasoning for the number of df applies for computing a Student's t-test or F-test for confidence intervals as well as when computing the coefficient of regression for simple versus multiple regression parameters. The reader will find the above equations reiterated when referring to classic statistical texts and when checking reliable university websites [6,7].

So where is the controversy in deciding df for any statistical parameter calculation when using multivariate statistics? The reader is referred to Table 117-1 for a sampling of reliable technical sources for the number of df to use for calculation of the various calibration and validation statistics used in multiple regression [1,3–5,8–17].

Table 117-1 Degrees of Freedom to Use for Calculation of the Various Calibration and Validation Statistics Used in Simple and Multiple Regression—cont'd

References	Reference Page(s)	Statistical Parameter	Degrees of Freedom (df)	Comment(s)
[14]	110	SEE	$n-2$	Minus 1df for slope and 1df for intercept (simple regression)
[15]	206, 207	df regression	n	Simple regression
	206, 207	df residuals	$n-k-1$	$n-2$ for simple regression
	260	df regression	m	Multiple regression
	260	df residuals	$n-m-1$	Multiple regression
[16]	94	SEE	$n-k-1$	k = Terms or PCR, PLS latent variables
	95	SEP (discussed)	$n-1$	k = Terms or PCR, PLS latent variables
[17]	Personal communication EAS	RMSECV	$n-1$ mc or n not mc	mc = Mean centered application
	Personal communication EAS	SEP	$n-k-1$ mc or $n-k$ not mc	k = Terms or PCR, PLS latent variables

[a] *For standards the actual page number of the standard is referred to, not the page number of a compendial version of a standard. For book references pages refer to specific editions of the book as outlined by date and in the reference section of this chapter.*
[b] *Note that df depends upon intended use, and one must always state the formulae used and correctly compute results for paired tests. Note that effective data rank depends upon use.*
[c] *Differing applications.*

GENERAL DISCUSSION

The discussion surrounding the issue of df is related to uniformity when comparing multiple experiments and evaluating the relative magnitude of errors derived from two or more methods. For example, when comparing a PLS model with a neural network or alternative MLR technique; thus, what does the magnitude of the residual errors indicate? The size of the residual error when comparing analytical methods is a powerful indicator of which method is optimized for the specific calibration experiment in question. So unless computational methods for error determination are identical, the size of the error calculations might differ significantly. So now that one knows to use the identical equation (and df) for comparative method studies, what equations and df should be used? Well there is the catch; in the spectroscopy quantitative analysis world, the use of empiricism has represented much of the applied technique over time. This approach indicates a loose application of precise theory in favor of selecting methods which appear to give optimum results for any particular calibration experiment case.

Chemometricians may have a problem with the issues associated with df, but statisticians do not find this issue to be problematic. For statisticians it is very clear: for all statistics, if you use the correct number of df then the expected value of the statistical parameter for a sample is identical to the population value for that corresponding parameter. For example, when computing simpler statistics such as mean and SD, the use of df are well defined as outlined earlier. The fact of the matter is, that in practice,

it hardly makes a difference. For example, the effect on a SD calculation of using n or $n-1$ is the square root of the ratio, so if n is, say, 10 or more, then the difference in the calculated value is $<1\%$. That is far less than the CI for that number of df. Since many samples are typically recommended for any spectroscopy quantitative calibration, approximately 100 or more, the difference is the square root of 100/99. Such differences represent less than the noise level of an instrument.

SUMMARY

Note then that the degrees of freedom for multivariate calibration depend upon intended use, and since there is some disagreement, when reporting results, one must always state the formula used for calculation and correctly compute the results. The effective data rank depends upon the application and when comparing methods and equations one should be certain to use the same equation for paired comparisons and clearly state what operations and formulae were used for all computations. As this subject gains more attention we will undoubtedly address additional issues related to it in the future.

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CHAPTER

BIAS AND SLOPE CORRECTION

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INTRODUCTION

The tedious requirement of continuous bias adjustment for every product and constituent calibration is primarily due to four factors. These factors are (1) reference laboratory differences, (2) the drift in product chemistry/spectroscopy requiring continuous updating of calibrations, (3) the drift or changes in spectral characteristics from a single spectrophotometer making measurements over time, and (4) the consistent differences in spectral characteristics between spectra measured from multiple spectrophotometers.

Reference laboratory differences are due to a true bias in the reference chemistry measurement values caused by different analysts, different laboratories, or fundamental differences in analysis methods. The drift in product chemistry/spectroscopy is caused by changes in raw materials, manufacturing processes, or drifting ecotype expression in natural products. This product drift is best accommodated for by performing recalibration rather than bias changes. However, if the user has robust and well-characterized calibration models with consistent and audited reference laboratory results, the primary cause of bias is brought about by within or between instrument differences in spectra and their resulting prediction results as compared to those spectra used for calibration. The key instrument factors requiring continuous biasing of prediction results are given above by factors (3) and (4).

The main issues associated with single instrument changes over time are minimal, mostly due to slight mechanical, electronic, or throughput (etendue) variations within a single instrument. These differences should be small and random for a properly functioning spectrophotometer. Also over time the spectral characteristics for instruments will slowly drift due to soiling of optical surfaces and lamp spectrum changes. The issues associated with multiple instruments involving calibration transfer from

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a primary instrument used for calibration to another "secondary" instrument or set of "secondary" instruments are more problematic. The differences between instruments cause most of the bias issues common to calibration transfer, namely (1) wavelength registration differences, (2) photometric offset, and (3) linewidth or spectral shape differences. *The bias for each product and constituent prediction is caused by the relatively consistent differences between spectra measured from different instruments. The bias represents a zero-order correction in an attempt to accommodate for one or several consistent differences between instruments.* As long as the differences between instruments are relatively consistent the bias changes will perform a very basic and functional correction for predicted values. If instrumental measurement differences within a single instrument over time or between different instruments can be mitigated, or even eliminated, the bias due to instrument differences would vanish. Under such an advanced technological approach the tedious work and difficulties associated with bias adjustments after calibration transfer could be eliminated for each product and constituent.

In our Chapter 113 covering this subject, also found in reference [1], we stated that instrumentation that is precisely alike in spectral shape from one instrument to another when measuring an identical sample will give precisely the same results using the same calibration equation across multiple instruments. We have discussed this issue in the past and demonstrated that when using NIR spectrophotometers for predictions, even if one uses random numbers for the dependent variables (representing the reference or primary laboratory results), different instruments will closely predict these random numbers across instruments following calibration transfer. Thus the instrument could be considered agnostic in terms of the numbers it generates using a spectral measurement when combined with multivariate calibration equations [2]. This is important and indicates that the precise spectral shape is the key feature determining the precise numbers reported as representative of a chemical or physical property of a sample (i.e., the prediction result). It is also important to note that the data structure used to develop a calibration (viz., spectra and reference values) will determine what exactly is reported as the chemical or physical properties of a sample.

We have also pointed out in the past some of the various requirements for successful calibration transfer and the wide differences between commercial NIR instrumentation in terms of agreement for wavelength and photometric axes registration [3–8]. And further we have reminded ourselves that spectrophotometers using Beer's law (and even Raman spectrometers that "self-determine" pathlength) measure moles per unit volume or mass per unit volume and do not measure, or rather track exactly, odd or contrived reference values simply by adding more terms to the regression calibration model [9,10]. Note that irrespective of the chemometric or algorithmic approaches used to develop a calibration model, an inconsistent spectral shape with changing X (wavelength) and Y (photometric) axes registrations over time, and between instruments, will disrupt any model such that bias correction must remain a requirement following calibration transfer for each product and constituent model.

ANALYSIS OF THE ISSUES

If one looks at the simple univariate case from reference [1] one can see there is a definite relationship between SEP, bias, and slope related to common changes in wavelength registration, photometric levels, and linewidth or spectral shape variation. These are the usual changes in instrumentation between different serial numbers of the same instrument design. One can actually establish a set of experiments for any product and constituent univariate or multivariate model to measure the

instrument differences. The analyte band absorbance for the example calibration data set ranged from 0.89 to 1.12 Au (representing that a small change in absorbance yields a relatively large change in concentration), and the original linewidth for the calibration set was 16.4 nm.

INSTRUMENT DIFFERENCES VERSUS SEP

For Fig. 118-1A–C we observe the changes in the SEP based on differences in instrument measurement parameters to include wavelength registration changes, photometric offset, and lineshape a.k.a. linewidth. In this first set of figures we see dramatic changes in SEP with the common differences expected between most modern instrument types.

One observes from Fig. 118-1A that the SEP exhibits large variation with wavelength changes of ±1.0 nm. For this example one would expect that some form of calibration transfer algorithm, such as PDS or another form of instrument standardization, would be required to compensate for the large changes in SEP relative to this magnitude of wavelength registration differences. Again one might perform an experiment plotting these figures for any specific calibration when altering the spectral data in order to measure the change in prediction values relative to differences in instrument measurement characteristics.

In Fig. 118-1B we observe the expected changes in prediction results as SEP with a simple offset change that would be of the magnitude expected between different instrument manufacturers. Note that this is a simple univariate case and some multivariate models may be developed to include some compensation or accommodation to an offset phenomena. In the case shown, a correction of the photometric axis for multiple instruments would be adequate to adjust the prediction results to be accurate.

In Fig. 118-1C one observes the effects on SEP of an increasing linewidth. The SEP continues to increase as the linewidth changes relative to the calibration spectral data. One might use several methods to "correct" the prediction results, such as bias, or slope, but making instruments alike with precise tracking of predicted results would require an actual bandwidth correction for the spectra.

INSTRUMENT DIFFERENCES VERSUS BIAS

For bias changes related to each type of instrument differences we look at the following. Changes in wavelength registration, offset, and linewidth have a dramatic effect on the bias of prediction results.

From Fig. 118-2A, we observe the wavelength effects on bias are dramatic and demonstrate that a ±1.0 nm shift will cause a bias change of approximately −0.9 for a constituent with a mean concentration of 15. A difference in wavelength registration of this magnitude is not out of the question when comparing different instruments from the various manufactures. Not surprisingly, a +0.5 nm shift indicated a bias change of approximately −0.7 concentration units.

Possibly even more surprising is the bias change caused by photometric differences that might be expected between instruments (Fig. 118-2B). For this sensitive model example, the changes of ±0.10 Au caused a bias change of approximately ±4.5 units for a calibration with an average concentration of 15. One must note that when a slight change in absorbance represents a large change in concentration for a constituent, the photometric accuracy is critical to accurate prediction results.

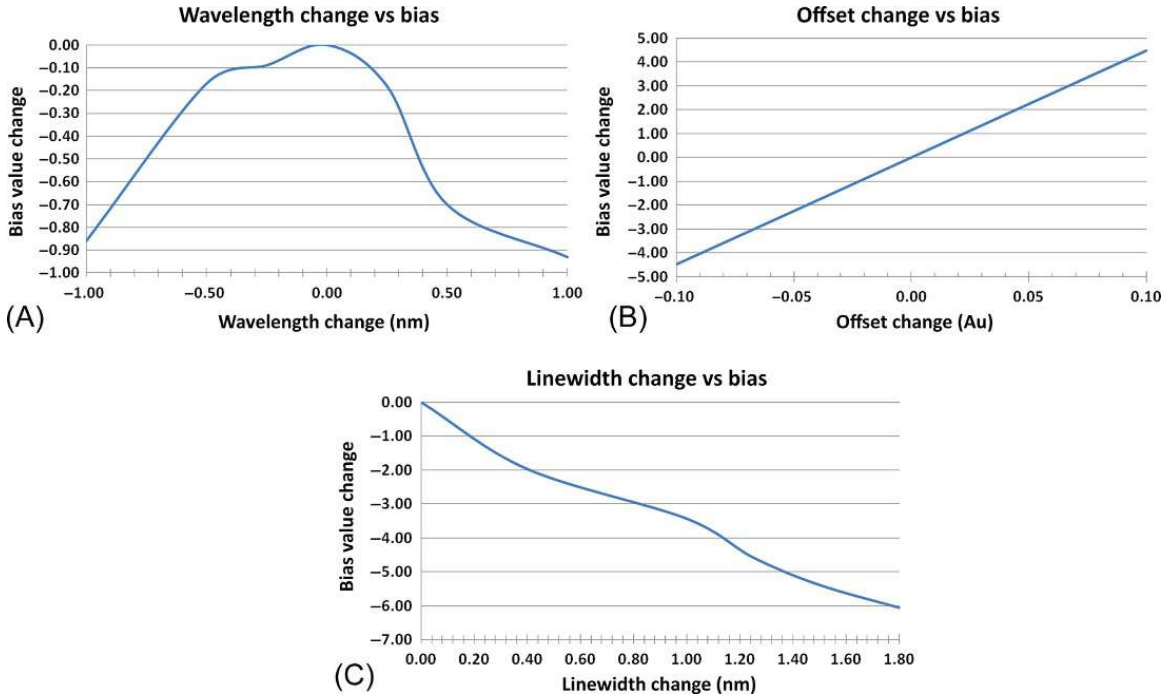


FIG. 118-2

(A) Changes in bias relative to spectral wavelength registration differences. (B) Changes in bias relative to spectral photometric offset differences. (C) Changes in bias relative to spectral linewidth differences.

In Fig. 118-2C one observes that a change in linewidth of +1.8 nm, where the original linewidth of spectra are changed from 16.4 to 18.2 nm, causes a bias of approximately −6.0 concentration units for a dataset averaging 15 concentration units.

INSTRUMENT DIFFERENCES VERSUS SLOPE

For prediction value slope change we may discover the following using this example. Offset has no effect on slope (Fig. 118-3B), but wavelength and linewidth changes do have a significant effect on slope. Wavelength is of much lesser effect on slope than linewidth (e.g., see Fig. 118-3A and C, respectively). Note that differences in wavelength registration of ±1.0 nm change slope minimally from 0.991 to 0.970 (Fig. 118-3A). However one may observe from Fig. 118-3C that a change in linewidth of +1.8 nm where the original linewidth of 16.4 nm is broadened to 18.2 nm, causes a slope change of from 1.0 (no change) to approximately 0.8 for this linewidth change of +1.8 nm.

SUMMARY

From this basic analysis of the effects of wavelength registration, photometric offset, and linewidth differences between instruments on the predicted values indicated by SEP, bias, and slope, one has

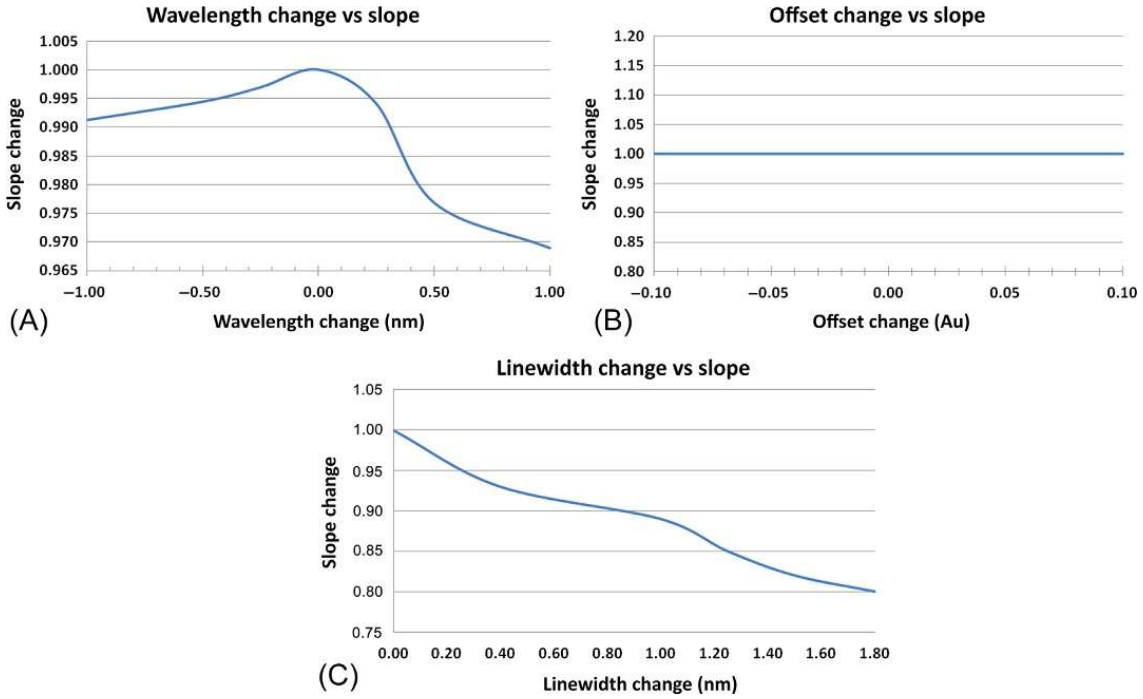


FIG. 118-3

(A) Changes in slope relative to spectral wavelength registration differences. (B) Changes in slope relative to spectral photometric offset differences. (C) Changes in slope relative to spectral linewidth differences.

a demonstrated method of determining the sensitivity of any product calibration to typical between instrument variation. One could easily write an automated software routine to alter spectral files and run predictions using any calibration model to determine that model's sensitivity to typical instrument differences.

In chapter 113 we discuss the statistical tests used for assessing the significance of SEP, bias, and slope in such detail as to describe the relationships between instrument differences required for spectral transfer to be completed seamlessly. In other words we answer the question of how close in spectral response instruments must be to successfully transfer calibrations without using standardization methods.

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